Migration Choice In Levels. These were simply

$$\pi^n_{ll',t} = \frac{\exp\left(\beta V^n_{l',t+1} - (\kappa_{ll'} + m_{ll',t})\right)^{\frac{1}{\nu^n}}}{\sum_{k\in\mathcal{L}} \exp\left(\beta V^n_{k,t+1} - (\kappa_{ll'} + m_{ll',t})\right)^{\frac{1}{\nu^n}}}$$

Let $u^n_{l,t} = \exp V^n_{l,t}$ and $\mu_{ll',t} = \exp m_{ll',t}$. The equilibrium system of equations we can use to solve the model without having to estimate bilateral flows comes from writing values and migration rates in changes.

Migration Rates in Changes

$$\begin{aligned}
\frac{\pi^n_{ll',t}}{\pi^n_{ll',t-1}} &= \frac{\exp(\beta V^n_{l',t+1} - (\kappa_{ll'} + m_{ll',t}))^{\frac{1}{\nu^n}}}{\exp(\beta V^n_{l',t} - (\kappa_{ll'} + m_{ll',t-1}))^{\frac{1}{\nu^n}}} \times \frac{\sum_{k\in\mathcal{L}} \exp\left(\beta V^n_{k,t} - (\kappa_{lk} + m_{lk,t-1})\right)^{\frac{1}{\nu^n}}}{\sum_{j\in\mathcal{L}} \exp\left(\beta V^n_{j,t+1} - (\kappa_{lj} + m_{lj,t})\right)^{\frac{1}{\nu^n}}} \\
&= \frac{(\dot{u}^n_{l',t+1})^{\frac{\beta}{\nu^n}} (\dot{\mu}_{ll',t})^{\frac{1}{\nu^n}}}{\sum_j \frac{\exp(\beta V^n_{j,t+1} - (\kappa_{lj} + m_{lj,t}))^{\frac{1}{\nu^n}}}{\sum_k \exp(\beta V^n_{k,t} - (\kappa_{lk} + m_{lk,t-1}))^{\frac{1}{\nu^n}}}} \\
\iff \pi^n_{ll',t} &= \frac{(\dot{u}^n_{l',t+1})^{\frac{\beta}{\nu^n}} (\dot{\mu}_{ll',t})^{\frac{1}{\nu^n}}}{\sum_j \pi^n_{lj,t-1} (\dot{u}^n_{j,t+1})^{\frac{\beta}{\nu^n}} (\dot{\mu}_{lj,t})^{\frac{1}{\nu^n}}} \pi^n_{ll',t-1}
\end{aligned}$$

EMAX in Changes.

Start by exponentiating the EMAX operator

$$\begin{aligned}
u^n_{l,t} &= w_{l,t} \left(\sum_{l'\in\mathcal{L}} \exp\left(\beta V^n_{l',t+1} - (\kappa_{l'l} + m_{l'l,t})\right)^{\frac{1}{\nu^n}} \right)^{\nu_n} \\
\iff \dot{u}^n_{l,t} &= \dot{w}_{l,t} \left(\sum_{l'\in\mathcal{L}} \pi^n_{ll',t-1} (\dot{u}^n_{l',t+1})^{\frac{\beta}{\nu_n}} (\dot{\mu}_{ll',t})^{\frac{1}{\nu_n}} \right)^{\nu_n}
\end{aligned}$$

Aggregate Labor Supply.

$$\begin{aligned}
L_{l,t} &= \left[\lambda^{1-\rho}_{l,t} (\alpha^F L^F_{l,t})^\rho + (1-\lambda_{l,t})^{1-\rho} (\alpha^D L^D_{l,t})^\rho \right]^{\frac{1}{\rho}} \\
&= \left[\lambda^{1-\rho}_{l,t} (\alpha^F \dot{L}^F_{l,t})^\rho (L^F_{l,t-1})^\rho + (1-\lambda_{l,t})^{1-\rho} (\alpha^D \dot{L}^D_{l,t})^\rho (L^D_{l,t-1})^\rho \right]^{\frac{1}{\rho}}
\end{aligned}$$

Aggregate Productivity.

Appendix of the paper has the derivation of Z with $z^D(\tau) = e^{\Gamma + \gamma^D \tau}$ and $z^F(\tau) = e^{\gamma^F \tau}$. This is equation A.11 from the paper

$$
\begin{aligned}
Z_{s,t}^{\frac{\rho}{1-\rho}} &= \int_{\Omega}\left[\int_0^{\mathcal{T}_{s,t}(\omega)} (\omega z^F(\tau))^{\frac{\rho}{1-\rho}} d\tau + \int_{\mathcal{T}_{s,t}(\omega)}^1 z^D(\tau)^{\frac{\rho}{1-\rho}} d\tau\right] dG(\omega) \\
&= \left(\frac{1-\rho}{\gamma^F \rho}\right)\left[\left(\frac{\alpha^F w_{s,t}^D}{\alpha^D w_{s,t}^F} \times \frac{1}{e^{\Gamma}}\right)^{\frac{\rho\gamma^F}{(1-\rho)(\gamma^D-\gamma^F)}} \mathbb{E}\omega^{\frac{\rho\gamma^D}{(1-\rho)(\gamma^D-\gamma^F)}} - \mathbb{E}\omega^{\frac{\rho}{1-\rho}}\right] \\
&+ e^{\frac{\rho}{1-\rho}\Gamma}\frac{1-\rho}{\rho\gamma^D}\left[e^{\frac{\rho\gamma^D}{1-\rho}} - \left(\frac{\alpha^F w_{s,t}^D}{\alpha^D w_{s,t}^F} \times \frac{1}{e^{\Gamma}}\right)^{\frac{\rho\gamma^D}{(1-\rho)(\gamma^D-\gamma^F)}} \mathbb{E}\omega^{\frac{\rho\gamma^D}{(1-\rho)(\gamma^D-\gamma^F)}}\right] \qquad (A.11)
\end{aligned}
$$

Resource Feasibility. Capital demand comes from standard Cobb-Douglas demand

$$
\begin{aligned}
(r+\delta)K_{l,t} &= \theta Y_{l,t} = \theta K_{l,t}^{\theta}(Z_{l,t}L_{l,t})^{1-\theta} \\
K_{l,t} &= \left(\frac{\theta}{r+\delta}\right)^{\frac{1}{1-\theta}} Z_{l,t}L_{l,t}
\end{aligned}
$$

where r, δ are exogenous. Then,

$$
\begin{aligned}
Y_{l,t} &= w_{l,t}^D L_{l,t}^D + w_{l,t}^F L_{l,t}^F + (r+\delta)\left(\frac{\theta}{r+\delta}\right)^{\frac{1}{1-\theta}} Z_{l,t}L_{l,t} \\
K_{l,t}^{\theta}(Z_{l,t}L_{l,t})^{1-\theta} &= w_{l,t}^D L_{l,t}^D + w_{l,t}^F L_{l,t}^F + (r+\delta)\left(\frac{\theta}{r+\delta}\right)^{\frac{1}{1-\theta}} Z_{l,t}L_{l,t} \\
\left(\frac{\theta}{r+\delta}\right)^{\frac{\theta}{1-\theta}} Z_{l,t}L_{l,t} &= w_{l,t}^D L_{l,t}^D + w_{l,t}^F L_{l,t}^F + (r+\delta)\left(\frac{\theta}{r+\delta}\right)^{\frac{1}{1-\theta}} Z_{l,t}L_{l,t} \\
\left(\frac{\theta}{r+\delta}\right)^{\frac{\theta}{1-\theta}} Z_{l,t}L_{l,t}\left[1-(r+\delta)\left(\frac{\theta}{r+\delta}\right)\right] &= w_{l,t}^D L_{l,t}^D + w_{l,t}^F L_{l,t}^F \\
\left(\frac{\theta}{r+\delta}\right)^{\frac{\theta}{1-\theta}} (1-\theta) Z_{l,t}L_{l,t} &= w_{l,t}^D L_{l,t}^D + w_{l,t}^F L_{l,t}^F
\end{aligned}
$$

Solving the Model

The algorithm for solving follows this follows.

0. Initialize

- Initialize some pre-period distribution of migration probabilities across space, $\pi_{ll',-1}^n$. When I turn to estimation, this will be taken from the data. For now just set them equal.
- Initialize time-zero stocks of labor across regions. Again, just go with uniform distribution.

1. Take a guess at the u-dots

Note. Ones are fine for this.

2. Calculate the pi's

3. Calculate the L's using the Law of Motion

$$L^n_{l,t+1} = \sum_{l' \in \mathcal{L}} \min\left\{\pi^n_{l'l,t} L^n_{l',t}, M^n_{l'l}\right\}$$

where M is a quota on flows from l' to l.

3. Solve the Static Equilibrium Problem

We have three equations in each location for this. The paper also introduces some functional forms. Let $z^D(\tau) = e^{\Gamma + \gamma^D \tau}$ and $z^F(\tau) = e^{\gamma^F \tau}$ where $\gamma^D > \gamma^F$ by comparative advantage. Combined with the Pareto distributed ω assumption this yields (appendix equation A.11)

$$\begin{aligned}
Z_{l,t}^{\frac{\rho}{1-\rho}} &= \int_\Omega \left[\int_0^{\mathcal{T}_{l,t}(\omega)} (\omega z^F(\tau))^{\frac{\rho}{1-\rho}} d\tau + \int_{\mathcal{T}_{l,t}(\omega)}^1 z^D(\tau)^{\frac{\rho}{1-\rho}} d\tau\right] dG(\omega) \\
&= \left(\frac{1-\rho}{\gamma^F \rho}\right)\left[\left(\frac{\alpha^F w^D_{l,t}}{\alpha^D w^F_{l,t}} \times \frac{1}{e^\Gamma}\right)^{\frac{\rho\gamma^F}{(1-\rho)(\gamma^D-\gamma^F)}} \mathbb{E}\omega^{\frac{\rho\gamma^D}{(1-\rho)(\gamma^D-\gamma^F)}} - \mathbb{E}\omega^{\frac{\rho}{1-\rho}}\right] \\
&+ e^{\frac{\rho}{1-\rho}\Gamma} \frac{1-\rho}{\rho\gamma^D}\left[e^{\frac{\rho\gamma^D}{1-\rho}} - \left(\frac{\alpha^F w^D_{l,t}}{\alpha^D w^F_{l,t}} \times \frac{1}{e^\Gamma}\right)^{\frac{\rho\gamma^D}{(1-\rho)(\gamma^D-\gamma^F)}} \mathbb{E}\omega^{\frac{\rho\gamma^D}{(1-\rho)(\gamma^D-\gamma^F)}}\right]
\end{aligned}$$

where we can see that

$$\int_\Omega \left[\int_0^{\mathcal{T}_{l,t}(\omega)} \omega z^F(\tau)^{\frac{\rho}{1-\rho}} d\tau\right] dG(\omega) := \left(\frac{1-\rho}{\gamma^F \rho}\right)\left[\left(\frac{\alpha^F w^D_{l,t}}{\alpha^D w^F_{l,t}} \times \frac{1}{e^\Gamma}\right)^{\frac{\rho\gamma^F}{(1-\rho)(\gamma^D-\gamma^F)}} \mathbb{E}\omega^{\frac{\rho\gamma^D}{(1-\rho)(\gamma^D-\gamma^F)}} - \mathbb{E}\omega^{\frac{\rho}{1-\rho}}\right]$$

and

$$\int_\Omega \left[\int_{\mathcal{T}_{l,t}(\omega)}^1 z^D(\tau)^{\frac{\rho}{1-\rho}} d\tau\right] dG(\omega) := e^{\frac{\rho}{1-\rho}\Gamma} \frac{1-\rho}{\rho\gamma^D}\left[e^{\frac{\rho\gamma^D}{1-\rho}} - \left(\frac{\alpha^F w^D_{l,t}}{\alpha^D w^F_{l,t}} \times \frac{1}{e^\Gamma}\right)^{\frac{\rho\gamma^D}{(1-\rho)(\gamma^D-\gamma^F)}} \mathbb{E}\omega^{\frac{\rho\gamma^D}{(1-\rho)(\gamma^D-\gamma^F)}}\right].$$

Also, the appendix of the paper calculates the two expectations involved

$$\mathbb{E}\omega^{\frac{\rho}{1-\rho}\frac{\gamma^D}{\gamma^D-\gamma^F}} = \frac{(\gamma^D-\gamma^F)(1-\rho)}{(\gamma^D-\gamma^F)(1-\rho) - \rho\gamma^D} \quad \text{and} \quad \mathbb{E}\omega^{\frac{\rho}{1-\rho}} = \frac{1-\rho}{1-2\rho}$$

These two integrals are important because they show up directly in the labor market clearing conditions, which, together with the resource feasibility condition pin down wages and task prices

$$\begin{aligned}
\left(\frac{w^D_{l,t}}{P_{l,t}}\right)^{-\frac{\rho}{1-\rho}} &= \left(\frac{L^D_{l,t}}{Q_{l,t}}\right)^\rho \left(\int_\Omega \int_{\mathcal{T}_{l,t}(\omega)}^1 (\alpha^D z^D(\tau))^{\frac{\rho}{1-\rho}} d\tau dG(\omega)\right)^{-\rho} \\
\left(\frac{w^F_{l,t}}{P_{l,t}}\right)^{-\frac{\rho}{1-\rho}} &= \left(\frac{L^F_{l,t}}{Q_{l,t}}\right)^\rho \left(\int_\Omega \int_0^{\mathcal{T}_{l,t}(\omega)} (\omega\alpha^F z^F(\tau))^{\frac{\rho}{1-\rho}} d\tau dG(\omega)\right)^{-\rho} \\
\left(\frac{\theta}{r+\delta}\right)^{\frac{\theta}{1-\theta}} (1-\theta) Z_{l,t} L_{l,t} &= w^D_{l,t} L^D_{l,t} + w^F_{l,t} L^F_{l,t}
\end{aligned}$$

where recall that

- $Z_{l,t} = \left(\int_\Omega \left[\int_0^{\mathcal{T}_\omega} (\omega z^F(\tau))^{\frac{\rho}{1-\rho}} d\tau + \int_{\mathcal{T}_\omega}^1 z^D(\tau)^{\frac{\rho}{1-\rho}} d\tau \right] dG(\omega) \right)^{\frac{1-\rho}{\rho}}$
- $L_{l,t} = \left[\lambda_{l,t}{}^{1-\rho} (\alpha^F L_{l,t}^F)^\rho + (1-\lambda_{l,t})^{1-\rho} (\alpha^D L_{l,t}^D)^\rho \right]^{\frac{1}{\rho}}$
- $\lambda_{l,t} = \int_\Omega \left[\int_0^{\mathcal{T}_\omega} (\omega z^F(\tau))^{\frac{\rho}{1-\rho}} d\tau \right] dG(\omega) / Z_{l,t}^{\frac{\rho}{1-\rho}}$

Note. $Q_{l,t} = Z_{l,t} L_{l,t}$ in equilibrium which should be in the code since the integrals also depend on the relative wages (i.e. don't simply solve over Q)

4. Update the u-dots

Update using the EMAX in changes and update the convergence var. If below tolerance done, else start 2. with updated u-dots.